

MODERN SENSORS For DEFENCE And MILITARY APPLICATIONS

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Security systems rely on smart and intelligent sensor technologies and, hence, sensors have become an essential part of military systems. This article covers some latest sensors used in military applications

Modern military and defence environments require proven, reliable and scalable technologies. Sensors are a critical part of the technologies as these provide solutions to the whole defence ecosystem, including complex controls, measurements, monitoring and execution. Military and defence systems include drones, spacecrafts, missiles, military vehicles, ships, marine systems, satellites and rockets. These systems work in the harshest of environments during normal as well as combat operations. Internal and external security systems rely on smart and intelligent sensor technology for surveillance, intelligence and combat operations. Thus, sensors have become an integral part of military systems and the performance needs of all military vehicles, equipment and related systems.

Sensors are used in flight controls, propulsion, environmental monitoring, weapon

controls, indicators, communications and so on. You can find sensors in various security systems, explosives detection systems, chemical warfare, crime detection systems, intrusion detection systems and civil establishments, among others. Sensors are also used in battlefield surveillance systems for the conduct of modern warfare. These are deployed on land, aerial platforms, in space and underwater, to keep a 24x7 watch over a particular targeted zone.

The electronic circuitry including various sensors is a critical part of the missile system. For example, Bloodhound missile was a relatively advanced missile for its era (the Cold War) with an advanced continuous-wave semi-active radar homing system, offering excellent performance against electronic countermeasures and fire control. With modern technology, especially in sensory systems, now the missiles are much more sophisticated, powerful and accurate as compared to traditional ones.

Some of the important sensors include advanced earth observation (EO) and infrared (IR) sensors, target and weapon seekers, laser sensors for detection and imaging of targets, radar sensors for synthetic aperture radar (SAR) and inverse SAR (ISAR), sensors for manned and unmanned aerial platforms, and electro-optical sensors to detect fog, smog and dust. For example, Cartosat-2C, India's latest EO satellite, can capture various images including trucks and army columns from space. It can be used to monitor the borders and internal regions for security purposes. Images

Fig. 1: Bloodhound missile (Credit: <https://commons.wikimedia.org>)

